

Leader learning loss function in neural network classification

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ABSTRACT

Deep learning, based on Empirical Risk Minimization (ERM), typically aims to fit the ideal outputs of all samples due to its large capacity. However, models trained based on empirical losses like cross entropy (CE) or mean square error (MSE), often learn unnecessary information during classification, leading to premature overfitting. On the other hand, the result-focused loss functions, i.e., zero-one loss or hinge loss, are hard to optimize and thus are rarely applied directly in neural network. This paper proposes a novel leader learning in classification, where CE is gradually trained by classification results using sample-dependent cost-sensitive learning. As complementary, the stepwise-changed CE covers the deficiency on classification error while preserving the advantage of fast convergence. In this way, the deviation between CE and classification error can be corrected. Experimental results demonstrate that the proposed leader learning has a more significant convergence trend than the baseline algorithms. Moreover, the loss function learned from a specific dataset has broad generality that can be transferred to other models as prior knowledge.

1. Introduction

With the rapid development of information technology, machine learning has received significant attention and has been applied to many complex tasks, such as image recognition [1,2] and natural language processing [3,4]. In these fields, deep learning is one of the most important sources of technological advances. The loss function employed in Empirical Risk Minimization (ERM) plays a pivotal role in neural network learning, guiding network optimization, and thus the loss function is also known as the objective function. The specific design of the loss function is an essential and important aspect in different tasks.

Classification is a fundamental task in machine learning. Distinguishing it from regression where outputs align with desired labels, classification models seek hyperplanes that maximize the separation between samples. Empirical losses, such as cross entropy (CE) and mean square error (MSE), are the most popular loss functions. CE measures the discrepancy between the probability distribution of the corresponding class and the desired distribution, typically computed via the softmax function, while MSE measures the Euclidean distance between the output and the expected label. Golik et al. in the literature [5] compared these two most popular loss functions and demonstrated that cross entropy has a faster convergence compared to mean square error that takes into account more complex optimization scenarios. However, Hui [6] empirically showed that CE outperforms MSE in image recognition, while in other domains like natural language processing,

the superiority of CE is not consistently observed. Furthermore, Daniel et al. showed that these two loss functions under over-parameterized models share the same property that long-standing training with tiny error can still enhance the implicit regularization [7]. Muthukumar et al. [8] made a similar conclusion in the over-parameterized model with continuous learning that all samples become support vectors, either CE or MSE.

Before the explosion of deep learning, zero-one loss and hinge loss served as prevalent loss functions in classification models, such as perceptron [9] and SVM [10]. However, the non-smooth property of these loss functions poses challenges in optimization of gradient descent [11, 12]. Presently, CE and MSE have become the default choices in various classification tasks due to their favorable convergence speed in gradient descent and good interpretability. However, it is also presented that the interpretation is somewhat confusing. Technically, MSE or CE aims to fit artificially assigned labels or ideal probability distributions. It is still substantially a fitting problem being approximated as a classification problem. Thus, there is a fundamental difference between fitting the ideal output and finding a separable decision boundary: the fitting based on empirical loss is a sufficient but unnecessary condition for learning a separable boundary. Since the large models always have an immense capability to achieve this sufficient condition, the target deviation between empirical loss and classification error is always readily ignored in deep learning. Inevitably, the target deviation between

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loss function and classification error makes the model learn something unnecessary, which may lead to overfitting, poor calibration or other problems during classification.

A technical effort on sample imbalance classification has been brought to our attention. Cost-sensitive learning, achieved by modifying the cost-sensitive factors in the loss function, addresses the challenge of overrepresented classes in data-driven models [13,14]. This approach has been widely applied in general sample imbalance classification and computer vision tasks [15,16]. Notably, some scattered findings provide valuable insights on classification tasks. For instance, a study [17] demonstrated that assigning different sensitivity coefficients to difficult-to-classify samples enhances the overall classification performance in unbalanced sample classification. Additionally, it is worth highlighting that the linear analytical solution for quadratic approximation of classification error is a special kind of weighted least squares, which coincides with the optimal solution for MSE in linear cost-sensitive learning [18].

In this paper, we propose a novel leader learning for neural network classification to address the intrinsic mismatch between CE and classification error. Our proposed method consists of two optimization phases in each epoch: model learning and loss function learning. During the model learning phase, we optimize the neural network using conventional gradient descent with general cross entropy (GeCE), where each sample is assigned a cost-sensitive factor in CE. In loss function learning phase, we train GeCE based on the zero-one loss. The cost-sensitive factors are adjusted based on classification results to keep the decision boundary close to misclassified samples. Throughout the optimization process, zero-one loss acting as a leader is the real objective function of the algorithm, while GeCE as an executor has a good convergence in terms of optimizing the neural network. The leader is to guide the model in finding the classification hyperplane, such as zero-one loss. The executor has good convergence in model training, such as MSE or CE. Furthermore, zero-one loss corrects the deviation between CE and classification error while CE covers the shortage of the non-smooth property of zero-one loss and retains its convergence advantage. In the experimental simulations, a visible synthetic dataset, ten real-life datasets, MNIST and CIFAR-10/100 are used to evaluate the proposed classification method. It is empirically demonstrated that the loss function learned from a specific dataset has broad generality, which can be transferred to other models as prior knowledge.

The rest of the paper is organized as follows. In Section 2, neural network classification and cost-sensitive learning are briefly described. The proposed leader learning, its algorithm and some analysis are presented in Section 3. Numerical results and performance comparisons are given in Section 4, and the conclusions are drawn in Section 5.

2. Preliminaries

This section discusses the traditional classification and cost-sensitive learning in neural network learning.

2.1. Neural network classification

Suppose $\{(x_i, y_i)\}_{i=1}^N$ is the given dataset where $x_i = [x_{i,1}, \dots, x_{i,n}]^T$ and $y_i = [y_{i,1}, \dots, y_{i,m}]^T$ where n indicates the number of input features and m indicates the number of categories. If the i th sample belongs to the c th category, $y_{i,c}$ is 1, and other labels are 0. $f_j(\cdot)$ represents the input-output mapping on the j th category. The class label can be predicted using the one-versus-all technique as

$$cls(f(x)) = \arg \max_{j \in \{1,2,\dots,m\}} f_j(x) \quad (1)$$

where $f = [f_1, f_2, \dots, f_m]^T$. The following two training criteria are usually considered.

- Mean square error (MSE):

$$l_{MSE} = \sum_{i=1}^N \|y_i - f(x_i)\|_2^2 \quad (2)$$

- Cross entropy (CE):

$$l_{CE} = \sum_{i=1}^N -\log p(x_i) \quad (3)$$

where $p(x_i) = \exp(f_c(x_i)) / \sum_{j=1}^m \exp(f_j(x_i))$.

Regarding CE as training criterion, softmax function converts the original model output into a probability distribution where each category output falls within $[0, 1]$, and the sum of all outputs is equal to 1. The interpretation of the probability distribution intuitively seems to be more acceptable than Euclidean distance in classification tasks. Moreover, CE exhibits a faster convergence speed and enables better generalization. Although CE is the most popular loss function in classification tasks, does it really make sense to use a probability distribution to explain the input-output mapping and use CE as the classification training criterion?

If we disregard the vague interpretation and focus solely on the input-output mapping, it can be concluded that CE is essentially as ideal probability fitting, where the category to which the sample belongs is fitted to 1 and the remaining categories are fitted to 0. According to (1), the goal of the classification is to identify the boundary that the probability of the correct category exceeds those of the other categories, rather than matching the specified output. In binary classification, maximizing likelihood is equivalent to maximizing accuracy [19]. However, in multi-classification, minimizing cross entropy or maximizing likelihood of true category cannot ensure maximum accuracy. For example, suppose there are two models for sample $y = [1 \ 0 \ 0 \ 0]^T$, with the predicted probability outputs $\hat{y}_{M_1} = [0.45 \ 0.55 \ 0 \ 0]^T$ and $\hat{y}_{M_2} = [0.4 \ 0.2 \ 0.2 \ 0.2]^T$, respectively. Based on CE, M_1 is better than M_2 , because it focuses only fitting expected probability distribution and $-\log 0.45 < -\log 0.4$. According to the one-versus-all technique, M_1 misclassifies the sample while M_2 classified the sample correctly. Actually, M_2 is a better classifier than M_1 .

Based on the aforementioned knowledge, it is evident that a clear deviation exists between these two tasks. In the strictest sense, fitting expected output by CE is a sufficient but unnecessary condition for learning a separable boundary, since gradient descent minimizes overall empirical risk, not classification error. Additionally, the dependence of classification accuracy on the density distribution of samples implies that poor sampling may lead to unfavorable classification performance, such as the issue of sample imbalance. Furthermore, the acquisition of unnecessary knowledge during training may contribute to premature overfitting.

On the other hand, zero-one loss and hinge loss that focus on the classification results are logical losses in classification tasks and have been widely applied in machine learning, as depicted in Fig. 1. Nevertheless, in deep learning, models with these loss functions are hard to optimize [11,12]. Therefore, although CE does not match the classification goal exactly in nature, it is still the most efficient loss function in neural network classification, yielding remarkable results. However, a large amount of recent improvement efforts about CE also illustrates the possibility of new training patterns. For instance, L-softmax [20] enhances the angular margin-based softmax by incorporating label information to improve class-wise geometric relationships. A-softmax [21] introduces a hyperparameter to control the degree of angular margin, thereby enhancing the discriminative power. Considering clustering of output features, center loss simultaneously learns a center for output features of each class and penalizes the distances between deep features and corresponding class centers [22]. Focal loss, which weights original cross entropy by a probability-dependent term, has garnered significant attention and has been applied in object detection and confidence calibration [15].

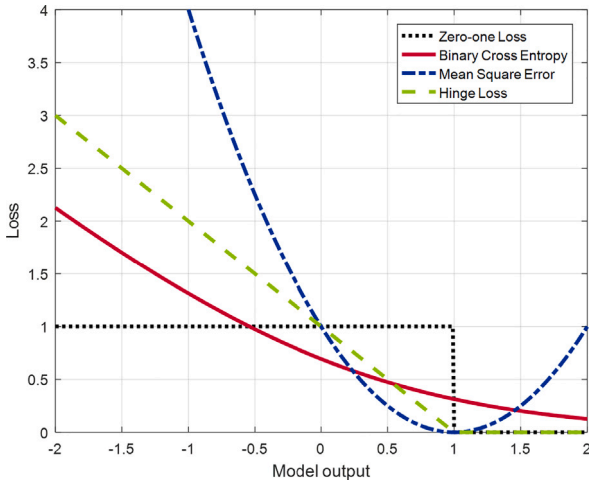


Fig. 1. Losses of different objective functions.

2.2. Cost-sensitive learning

Sample imbalance is a common problem in classification, where the statistical-driven models tend to oversee the overrepresented majority and neglect the underrepresented minority. Therefore, imbalanced class distribution poses a challenge for models based on ERM to learn the discriminative boundaries between the majority and minority classes. Previous attempts to handle class imbalance can be divided into two categories: data-level and algorithmic-level approaches. Data-level approaches re-balance the class distribution of the original imbalanced samples through random oversampling or random undersampling [23–25]. However, these methods often require complex data preprocessing and involve repeated sampling. On the other hand, algorithmic-level approaches, such as cost-sensitive learning, reinforce the model sensitivity towards the minority class by considering unequal misclassification costs. This method is widely employed in neural network classification to address imbalanced sample learning. In [26], a joint objective function was designed by combining the losses of different classes linearly. To avoid complex expert judgment in constructing cost matrix, Khan et al. [27] optimized both the model and cost parameters of different classes synchronously. Fernando et al. [17] proposed a dynamical weighted balanced loss not only for the imbalanced class but also for the convergence of challenging samples. In addition to the imbalanced class problem, cost-sensitive learning is also used to define the importance of different categories of classification in specific situations, such as privacy protection and medical diagnosis [28].

All of the above cost sensitivities are designed for class imbalance problems or specific tasks. However, some fragmented conclusions and experiments have implied that the sensitivity coefficient should focus not only on imbalanced sample classification but also on more general classification. In linear classification, the deterministic solution by quadratic approximation is a special form of weighted least squares, which can be viewed as a form of cost-sensitive learning for MSE [18]. Fernando et al. [17] also found that cost-sensitive learning for difficult samples can improve classification performance. Chai et al. proposed an adaptive re-weight method to tackle fairness issue [29]. Zhong et al. reweighted examples in the multi-label classification problem by considering instance complexities [30]. In [31], an adversarial training framework was presented to reweight the loss associated with individual training sample to improve robust generalization. Assigning the data-dependent weights is also usually discussed as boosting, a common machine learning framework that combines a number of weak classifiers into one strong classifier [32]. Other related but not same work about cost-sensitive learning can be found in concepts such as hard mining strategy [33], self-paced learning [34].

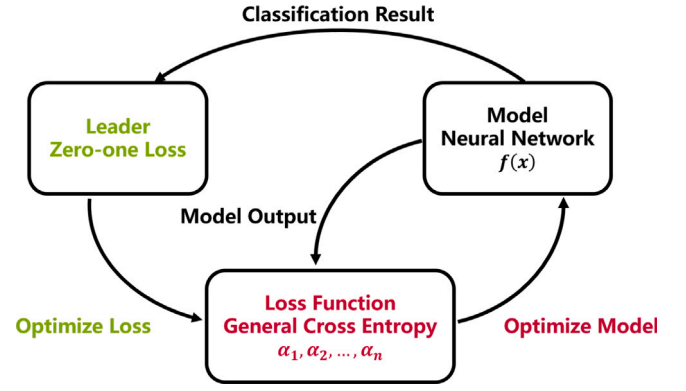


Fig. 2. Leader learning mechanism.

3. Leader learning

In the previous section, we discussed that CE or MSE is not inherently optimal for training neural network classifiers, and classification error is not suitable for deep learning. Given this consideration, it becomes imperative to devise an effective framework for optimizing neural network classifiers with the purpose of finding separable boundaries like zero-one loss and having good convergence and generalization like CE. In this section, we propose a new classification algorithm using loss function learning through another loss function in neural network training.

3.1. Leader learning loss function

In the rest of the paper, the analysis and discussion predominantly utilize CE, and it is worth to point out that MSE can also be consistent with the following findings. Unlike some work that specified a sensitivity factor for every category, each sample is set a sensitive factor in loss function. To differentiate our work from the previous efforts, the cross entropy with a general cost-sensitive factor per sample is named general cross entropy (GeCE). In our algorithm, the cost-sensitive factor in GeCE is learned by zero-one loss while GeCE directly optimizes the neural network model. The proposed learning algorithm with the loss function learning is called leader learning. In this framework, zero-one loss is the leader in optimizing GeCE so that GeCE as the executor can focus on classification results. The computation of GeCE can be described as follows.

$$l_{GeCE} = \sum_{i=1}^N -\alpha_i \log p(x_i) \quad (4)$$

where $\alpha_1, \alpha_2, \dots, \alpha_N$ are the parameters learned based on the following zero-one loss criterion.

$$l_{zero-one} = \begin{cases} 0 & \text{if } c_i = cls(f(x_i)) \\ 1 & \text{if } c_i \neq cls(f(x_i)) \end{cases} \quad (5)$$

where c_i and $cls(f(x_i))$ represent the true class and the predicted class of the sample x_i . The proposed leader learning is described in Fig. 2.

During each epoch, the cost-sensitive factor of the misclassified sample is magnified, which makes the model more sensitive to the error of the misclassified samples and forces the classification boundary close to the wrong samples. The update of cost-sensitive factor is given in (6).

$$\alpha_{mis,i} = \alpha_{mis,i} + \epsilon \quad (6)$$

where $\alpha_{mis,i}$ is the cost-sensitive factor of the misclassified sample $x_{mis,i}$, $i = 1, 2, \dots, k$; ϵ is the learning rate in loss function learning and k is the number of misclassified samples.

Although CE has good properties in numerous domains, it is still not equivalent to the classification error and is only a sufficient but not necessary condition for finding separable boundaries. With the ongoing growth of parameters in deep learning, current models always have enough ability to fit the ideal probability distribution. So this aspect often goes overlooked in deep learning. Nonetheless, it should be noted that this kind of loss function means that the neural network learns something unnecessary for the classification task. As a result, it may lead to a series of potential problems.

By introducing a dynamic leader learning into GeCE, the general optimization is not only based on fitting probability distribution but also focusing on separable boundary. The classification boundary can be forced close to the misclassified samples until the classification is completed. Moreover, cost-sensitive learning speeds up the convergence and changes the dynamic loss function by learning the cost-sensitive factor per sample. Notably, zero-one loss rectifies the mismatch between the CE and classification error, and in the meanwhile, CE compensates for the non-smooth property of zero-one loss and retains its convergence and generalization advantage.

Similar to our work, boosting is a widely adopted framework in machine learning that facilitates the dynamic adjustment of sample weights [32]. This framework primarily focuses on combining multiple weak classifiers into a single strong classifier, with fixed weights during the training of a single classifier. Moreover, Power Error Loss (PEL) provides a superior fit to the error signal distribution by effectively controlling the weighting of small and large output errors [19]. It utilizes a hand-crafted greedy schedule to adapt error exponents in each training step, allowing for the dynamic adjustment of the output error distribution [35]. In comparison, we propose leader learning, a method that dynamically learns sample weights while training a single deep model, with the objective of achieving an enhanced model and an optimized set of sample weights.

3.2. Algorithm description

At each epoch, the optimization process involves two stages: optimizing model parameters and optimizing cost-sensitive factors. These stages necessitate distinct optimization strategies and parameters.

Loss function learning. In loss function learning, the number of parameters is the same as the number of samples N . While it may seem logical to assign different learning rates to misclassified samples, such as using a larger learning rate for more challenging instances, manually adjusting numerous hyperparameters for different learning rates is impractical. To address this, we introduce the idea of momentum into the algorithm design so that the cost-sensitive coefficients of those difficult samples can also be learned effectively by the accumulation of momentum, as shown in the following.

$$\alpha_{mis,i} = \alpha_{mis,i} + v_{mis,i} \quad (7)$$

$$v_{mis,i} = v_{mis,i} \beta + \varepsilon \quad (8)$$

where ε is learning rate in (6); $0 < \beta < 1$ is the momentum decay coefficient and $v_{mis,i}$ is momentum. The learning rate ε is sensitive to balancing neural network learning and loss function learning. In general, a small learning rate causes leader learning to converge slowly, although a small learning rate is always effective. A larger learning rate may produce fluctuations or even non-convergence. If the sample is misclassified consecutively in the maximum t epochs, then its momentum in loss function learning can reach a maximum as in (9) and the momentum of all samples satisfies the inequality (10).

$$v_{max} = \varepsilon \beta^t + \varepsilon \beta^{t-1} + \dots + \varepsilon \beta + \varepsilon \quad (9)$$

$$\varepsilon \leq v_{mis,i} \leq \varepsilon \frac{1 - \beta^t}{1 - \beta} \quad (10)$$

With momentum optimization, the learning rate of consistently misclassified samples follows a cumulative sequence. The upper boundary of the learning rate is determined by the initial learning rate and the decay coefficient as depicted in (9) and (10). To simplify learning and focus only on misclassified samples, considering the classification of the sample that changes from incorrect to correct, its cost-sensitive factor will not be updated until it is misclassified again, and its momentum will be zeroed. Furthermore, an intriguing empirical observation will be presented in the following section, indicating that the learned GeCE for a specific dataset in a model has broad generality and can be transferred to improve another model as prior knowledge.

Model learning. Let the output of the model without softmax function be denoted by $z_{i,j}$, where $j \in \{1, \dots, m\}$ represents the j th class and $i \in \{1, \dots, N\}$ represents the i th sample. The normalized probability can be expressed as

$$p_{i,j} = \frac{e^{z_{i,j}}}{e^{z_{i,1}} + \dots + e^{z_{i,m}}} \quad (11)$$

$$\begin{aligned} \nabla_{z_{i,j}} l_{GeCE} &= \nabla_{z_{i,j}} \alpha_i \left(-z_{i,c} + \log \sum_j e^{z_{i,j}} \right) \\ &= \alpha_i \left(\frac{1}{\sum_j e^{z_{i,j}}} \nabla_{z_{i,j}} \sum_j e^{z_{i,j}} - \nabla_{z_{i,j}} z_{i,c} \right) \\ &= \alpha_i \left(p_{i,j} - \nabla_{z_{i,j}} z_{i,c} \right) \\ &= \alpha_i \left(p_{i,j} - \mathbf{I}(j = c) \right) \end{aligned} \quad (12)$$

where

$$\mathbf{I}(j = c) = \begin{cases} 1 & j = c \\ 0 & \text{otherwise} \end{cases}$$

It is easy to verify that GeCE becomes CE when the cost-sensitive factor of per sample is equal to 1.

3.3. Some analysis

Definition 1 (Local Minima). Consider continuous function f defined with domain D . Then f is said to have a local minimum at an interior point $\theta^* \in D$ if exists a real number $\delta > 0$ such that

$$f(\theta) \geq f(\theta^*), \forall \theta \in D \text{ satisfying } \|\theta - \theta^*\| < \delta \quad (13)$$

Lemma 1 (Gradient in Local Minima). If continuous function f have a local minimum at the point θ^* , $\frac{\partial f}{\partial \theta} = 0$.

We can obtain a sufficient convergence condition $0 < \varepsilon \leq \frac{2}{\eta L}$ for loss learning rate ε in Theorem 1 under assumption that the cost-sensitive coefficient of single sample is adjusted when optimization is trapped into local minima. According Theorem 1, the probability of all samples can be calibrated by adjusting the sample-dependent weight every time optimization is trapped into local minima. However, it takes significant computing resources to maintain the optimization into local minima. Leader learning is a compromise between computational complexity and stable learning.

Theorem 1. Considering second order derivable mapping $f(x, \theta)$ with optimization problem

$$\min_{\theta} \sum_{i=1}^n a_i l_i^t$$

where $l_i^t = -\log p_i^t$ and $p_i^t = \exp(f_c(x_i, \theta^t)) / \sum_{j=1}^m f_j(x_i, \theta^t)$ represents the probability output for sample x_i at t th epoch. The gradient of $l(\theta)$ satisfies L -Lipschitz constraint. Optimization is trapped into local minima at the point θ^t . For $0 < \varepsilon \leq \frac{2}{\eta L}$, update $a_k^* = a_k + \varepsilon$ and $\theta^{t+1} = \theta^t - \eta \frac{\partial \sum_{i=1}^n a_i l_i^t}{\partial \theta} - \varepsilon \eta \frac{\partial l_k^t}{\partial \theta}$, $\eta > 0$, there must be

$$p_k^{t+1} \geq p_k^t \quad (14)$$

Table 1
UCI machine learning datasets.

Dataset	Instance	Class	Feature	Training	Testing
Iris	270	3	4	150	120
Seeds	210	3	7	147	63
Wine	178	3	13	134	44
Wdbc	570	2	30	427	143
Thyroid	215	3	5	172	43
Cancer	683	2	9	546	137
Semeion	1593	10	256	1273	320
Image Segment	2310	19	7	2100	210
Robot Navigation	5456	24	4	4356	1100
Crowd Sourced Mapping	10545	27	6	8436	2109

Proof. According to update step $a_k^* = a_k + \varepsilon$, we can obtain

$$\theta^t - \theta^{t+1} = \eta \frac{\partial \sum_{i=1}^n a_i l_i^t}{\partial \theta} + \varepsilon \eta \frac{\partial l_k^t}{\partial \theta} \quad (15)$$

According to Definition 1 of local minima and Lemma 1 of gradient in local minima $\frac{\partial \sum_{i=1}^n a_i l_i^t}{\partial \theta} = 0$, we can obtain

$$\theta^t - \theta^{t+1} = \varepsilon \eta \frac{\partial l_k^t}{\partial \theta} \quad (16)$$

For a second-order differentiable objective function l_k^t that satisfies L -Lipschitz constraint $\left\| \frac{\partial^2 l_k^t}{\partial \theta^2} \right\| \leq L$ [36], we can have the following inequality

$$l_k^{t+1}(\theta^{t+1}) \leq l_k^t(\theta^t) + \left\langle \frac{\partial l_k^t}{\partial \theta}, \theta^t - \theta^{t+1} \right\rangle + \frac{L}{2} \|\theta^t - \theta^{t+1}\|_2^2 \quad (17)$$

The gradient descent is applied to optimization. Substituting (16) into (17), we obtain

$$l_k^{t+1}(\theta^{t+1}) \leq l_k^t(\theta^t) + \left(\frac{L\varepsilon^2\eta^2}{2} - \varepsilon\eta \right) \left\| \frac{\partial l_k^t}{\partial \theta} \right\|_2^2 \quad (18)$$

So, $\frac{L\varepsilon^2\eta^2}{2} - \varepsilon\eta < 0$ is sufficient condition for loss decline at each iteration. Because $0 < \varepsilon \leq \frac{2}{\eta L}$, we can obtain

$$l_k^{t+1}(\theta^{t+1}) \leq l_k^t(\theta^t)$$

and

$$p_k^{t+1} \geq p_k^t \quad \square$$

Corollary 1. Considering momentum optimization (7) and (8) for sample-dependent weight, Theorem 1 still holds when $\varepsilon \leq \frac{2-2\beta}{L\eta}$.

Proof. According to the sum of the proportional series on (9) and (10), we can get $v < \varepsilon \frac{1-\beta^t}{1-\beta} < \varepsilon \frac{1}{1-\beta}$. According to (7) and Theorem 1, $p_k^{t+1} \geq p_k^t$ hold when $v < \frac{2}{\eta L}$. Then we can obtain the sufficient condition $\frac{\varepsilon}{1-\beta} \leq \frac{2}{\eta L}$. $\square$

Theorem 1 and Corollary 1 consider overall optimization for all samples at local minima. Consider single sample optimization with objective $\alpha_i l_i^t$ that satisfies $\left\| \frac{\partial^2 l_i^t}{\partial \theta^2} \right\| \leq L$, we can obtain

$$l_k^{t+1}(\theta^{t+1}) \leq l_k^t(\theta^t) + \left(\frac{L\alpha_i^2\eta^2}{2} - \alpha_i\eta \right) \left\| \frac{\partial l_k^t}{\partial \theta} \right\|_2^2 \quad (19)$$

where α_i keeps increasing if x_i keeps getting misclassified in algorithm. However, η is fixed. So, gradient descent may not converge if α_i keeps increasing and $\frac{L\alpha_i^2\eta^2}{2} - \alpha_i\eta \geq 0$. On the basis of the above analysis, learning step and momentum in objective learning should be set small to maintain stable optimization. Besides, sample-dependent weights need to be cropped and $\alpha_k = \min \left\{ \alpha_k, \frac{2}{\eta L} \right\}$ if weight exceeds a certain threshold.

Table 2
Number of the nodes on the UCI.

Dataset	Layer1	Layer2	Learning Rate (*1e-3)
Iris	5	3	0.1
Seeds	10	3	0.1
Wine	10	3	1.0
Wdbc	10	2	0.1
Thyroid	10	3	1.0
Cancer	10	2	1.0
Semeion	10	10	1.0
Image Segment	50	19	0.05
Robot Navigation	50	24	0.5
Crowd Sourced Mapping	50	27	0.1

4. Numerical results

In this section, an extensive range of experiments are carried out to assess the effectiveness of the proposed leader learning. The experiments cover a synthetic dataset [37], ten publicly available datasets, as well as MNIST and CIFAR-10/100. We compare the performance of fully connected neural networks (FCNN) and different convolution neural networks (CNN) using CE, GeCE with leader learning, and other loss functions.

4.1. Double moons classification

In this subsection, 100 positive and 100 negative samples are randomly assigned based on the classic distribution of two moons. The parameter configuration is as follows. The number of nodes in the hidden layer is set to 5, and the number of output features is set to 2. The model is optimized using batch gradient descent with learning rate 0.01 and momentum 0.9. The learning rate and decay coefficient in loss function learning are $5e-3$ and 0.8. The activation function is sigmoid function. Half of the samples are selected in each batch. The experimental results of all values are averaged over 50 trials.

We use GeCE to represent the proposed leader learning and CE to represent the baseline algorithm with cross entropy loss in the following experiments. The decision boundaries of the two algorithms at 500 epochs are drawn in Fig. 3. The fitting ability of CE in classification is always based on the sample distribution since CE is concerned with probabilistic fitting rather than classification results. When the relative distance between two class sets becomes very close, CE is usually very hard to separate them, as shown in Fig. 3b. By dynamically learning GeCE based on the zero-one loss, the model is optimized with the overall goal on finding the separable hyperplane, so a separable decision boundary can be realized easily, as shown in Fig. 3a.

Furthermore, the convergence performance of two algorithms is plotted in Fig. 4 that GeCE has a faster convergence speed than CE. Then the performance with different learning rates in loss function learning is also shown in Fig. 4. It can be concluded that a small learning rate of the loss function learning is always effective, and the convergence of leader learning is sensitive to the learning rate. A large learning rate can speed up the convergence but also lead to the instability or even non-convergence of the algorithm. As analyzed in Section 3.3, large learning may lead to non-convergence of the algorithm.

4.2. Real-life datasets

There are ten real-life datasets from UCI machine learning datasets. The description of each dataset is listed in Table 1. The nodes of the model and the learning rate of loss function learning on each dataset are given in Table 2. Other parameters are the same in each dataset. Learning rate and momentum in model learning are set as 0.001 and 0.9. Momentum decay coefficient in loss function learning is 0.8. The

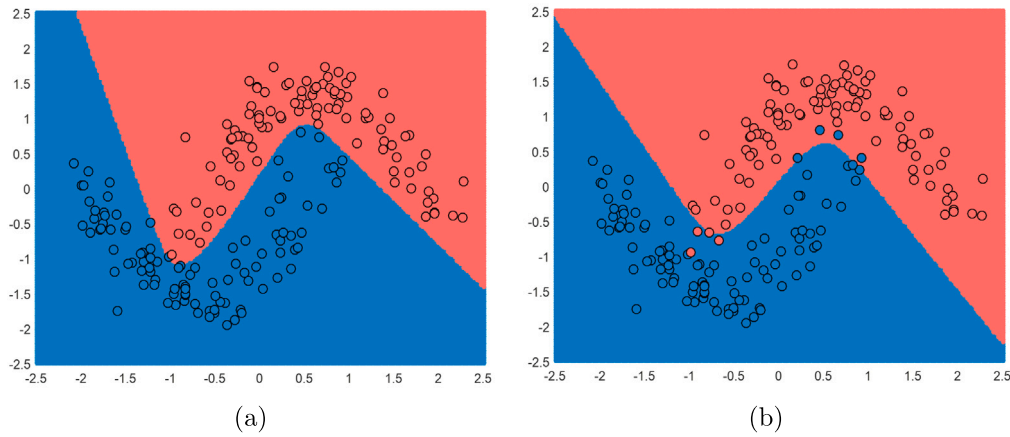


Fig. 3. Decision boundaries in double moons classification: a. GeCE b. CE.

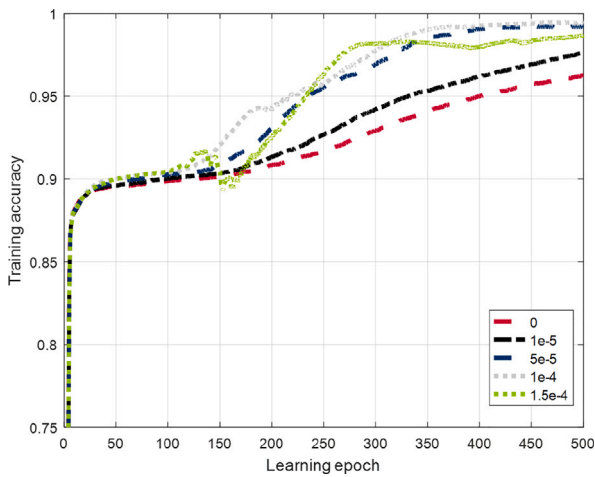


Fig. 4. Accuracy (%) of leader learning with different learning rates. 0 learning rate represents original CE.

maximum number of epochs is set to 500. The experimental results of all values are obtained over 20 trials.

The training accuracy and testing accuracy with different learning strategies are presented in Table 3. The number of samples of the different classes in all datasets is almost the same. By dynamic adjusting cost-sensitive factors in cross entropy based on zero-one loss, GeCE, which combines advantages of zero-one loss and CE, finds separable hyperplane and keeps excellent convergence speed. With an appropriate learning rate in loss function learning, GeCE can have better classification ability rather than CE. However, similar to the model learning, the determination of the learning rate in leader learning remains a challenge. Coordinating the learning rates of both processes requires substantial experience. In the hyperparameter adjustment, we initially modify the learning rate of the model learning, assigning a learning rate of 0 for the loss function learning. The optimal set of hyperparameters obtained through the adjustment is chosen as the experimental baseline. Subsequently, while keeping the model learning hyperparameters fixed, we gradually increase the learning rate from 0 to identify a parameter set that optimizes the sample weights. Although a very small learning rate for loss learning consistently yields better results, slightly larger learning rates can lead to oscillations of optimization.

In addition to CE, we did another experiment that leader learning can also improve the classification performance of MSE as an executor. The experimental results further support the notion that cost-sensitive

Table 3
Accuracy (%) on the UCI database.

Dataset	CE		GeCE	
	Training	Testing	Training	Testing
Iris	91.3	87.0	94.8	90.0
Seeds	97.8	76.3	98.6	76.8
Wine	96.8	95.7	99.1	96.8
Wdbc	97.2	98.3	97.3	98.2
Thyroid	79.2	73.6	98.8	92.3
Cancer	96.9	98.9	96.9	99.3
Semeion	96.9	82.8	99.7	84.1
Image Segment	93.3	92.9	93.9	93.3
Robot Navigation	91.0	87.8	93.1	88.2
Crowd Sourced Mapping	88.9	82.2	90.8	83.8

Table 4
Accuracy (%) on MNIST.

	CE	GeCE	GeCE+
Training accuracy	93.10	99.73	98.73
Testing accuracy	93.16	97.39	96.70

learning can enhance the model fitting capability, not only in scenarios involving sample imbalance problem but also in general classification tasks.

4.3. MNIST

The MNIST database of handwritten digits has a training set of 60,000 samples and a test set of 10,000 samples. The digits have been size-normalized and centered in a fixed-size 28×28 image. In this subsection, a three-layer fully connected neural network is tested on this database. The node of three layers are set to 50, 30 and 10, respectively. The learning rates in the two learning phases are 0.001 and 0.0001. The momentum and the momentum decay coefficient in the two learning phases are 0.9 and 0.8, respectively. Weight decay is 0.0001. The batch size is set to 100. GeCE+ represents the model with fixed GeCE learned from another model with leader learning. The experimental results of all values are obtained over 10 trials (see Table 4).

As shown in Fig. 5, GeCE can still have a significant convergence trend while CE can no longer achieve better convergence accuracy after a few iterations. At 150 epochs, GeCE can realize 99.73% training accuracy and 97.39% testing accuracy, which are improved by 6.63% and 4.23% respectively relative to the baseline CE loss. From the convergence trend, GeCE can still improve training accuracy and eventually find a separable hyperplane.

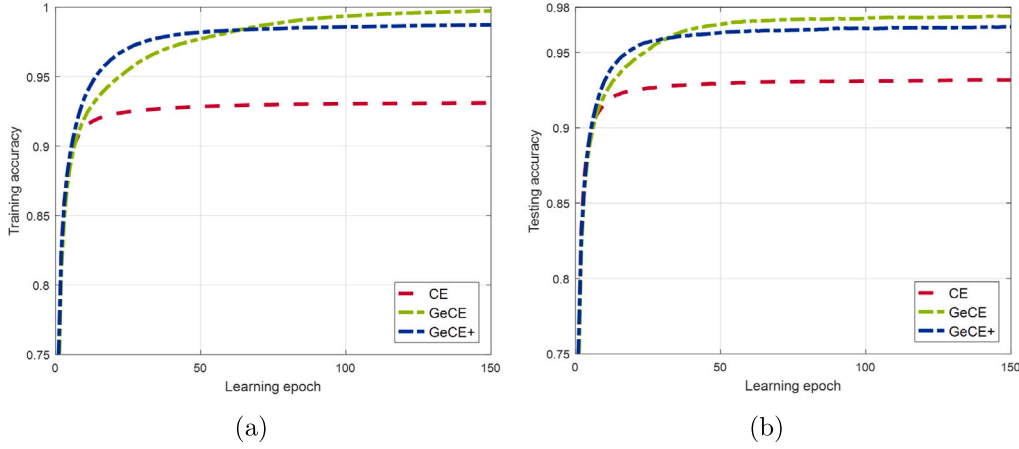


Fig. 5. Accuracy (%) on MNIST: a. Training accuracy b. Testing accuracy.

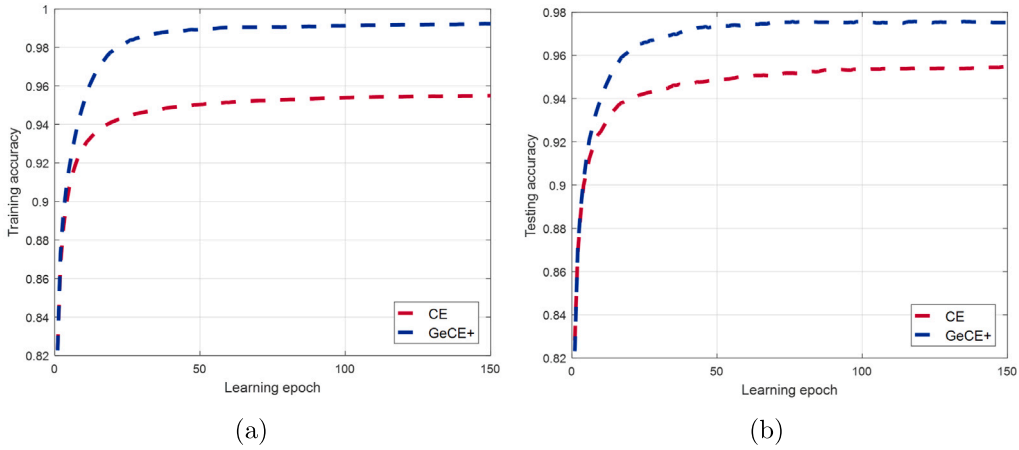


Fig. 6. Accuracy (%) between CE and GeCE+ with fixed loss function learned from different structures: a. Training accuracy b. Testing accuracy.

The underlying cause for this phenomenon lies in the change in the objective of model training. Fitting output is a more difficult task demanding a higher-capacity network than finding the classification boundaries. Furthermore, biased tasks constrain the model capacity in its original objectives and make the model learn something unnecessary. We posit that the incomplete match between classification error and CE contributes to the overfitting. Since it affects the generality and robustness of the model, the above target deviation will inevitably cause problems in deep learning, which need to be further discovered.

Besides, we present experimental results illustrating the transferability of the learned GeCE to another model with a different initialization but identical structure, as depicted in Fig. 5. GeCE+ with a fixed general loss function can realize a faster convergence rate than GeCE at the beginning but no better accuracy at the end of training. Moreover, we conduct another simulation, as depicted in Fig. 6, where the learned GeCE is applied to a different structure with 200 hidden nodes in the first layer and 10 nodes in the output layer. The results demonstrate that incorporating the learned cost-sensitive factors as prior knowledge can enhance the model performance. Meaningfully, the learned cost-sensitive factors for a specific database have generality for neural network learning.

4.4. CIFAR-10/100

In this subsection, leader learning is evaluated on different CNNs compared with different loss functions under datasets CIFAR-10/100. The first experiment is performed on a relatively shallow LeNet. In this

experiment, the learning rate of model learning is 0.01. Weight decay is $1e-4$. The learning rate and momentum coefficient in loss function learning are $1e-5$ and 0.9. The batch size is 100.

The experimental results are shown in Fig. 7 that GeCE has better performance than baseline algorithm CE. Surprisingly, GeCE+ with the fixed learned loss function even realizes higher training accuracy and testing accuracy than baseline which means the learned cost-sensitive factors for CIFAR-10 have good robustness and generality.

Since the capacity of deep model is large enough to achieve zero classification error, the effect of plain zero-one loss is not significant in the over-parameterized models. To verify leader learning in over-parameterized models, we use zero-one loss with hard probability margin instead of the plain zero-one loss in shallow models.

$$l_{zero-one} = \begin{cases} 0 & \text{if } p(x_i) < \sigma \\ 1 & \text{if } p(x_i) \geq \sigma \end{cases} \quad (20)$$

where p denotes the probability output belonging to true category [38]. The experimental results compared with A-softmax [21], LM-softmax [39], focal loss [15] and center loss [22] in ResNet17 [40], ResNet35 [40], VGG16 [41] and MobileNetV1 [42] on CIFAR-10/100 are presented in Tables 5 and 6. In the following experiments, margin σ is set to 0.9, the learning rate of model is 0.1 and decays to ten percent at the 50% and 75% epochs. As can be seen from Tables 5 and 6, our proposed method is still effective. Although such a method has no advantage over other advanced loss functions, it is still a significant improvement over the baseline. We guess that complex iterative adjustment for sample weights hinders gradient optimization

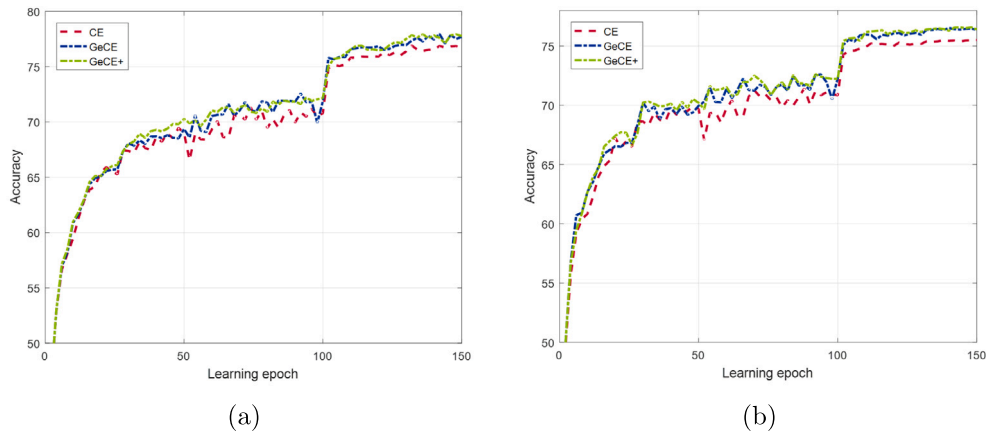


Fig. 7. Accuracy (%) of CIFAR-10 on LeNet: a. Training accuracy b. Testing accuracy.

Table 5

Top-1 accuracy (%) of CIFAR-10 on over-parameterized models.

Algorithms	ResNet17	ResNet35	VGG16	MobileNetV1
Cross entropy	92.1	93.0	91.8	90.8
Ours	92.7	94.0	92.5	91.5
A-softmax	93.2	94.2	92.9	91.7
LM-softmax	92.2	93.6	92.0	90.9
Focal loss	92.6	93.9	92.7	91.4
Center loss	92.7	93.8	92.4	91.6

Table 6

Top-1 accuracy (%) of CIFAR-100 on over-parameterized models.

Algorithms	ResNet17	ResNet35	VGG16	MobileNetV1
Cross entropy	73.2	74.2	72.6	70.2
Ours	74.1	74.8	73.2	70.9
A-softmax	74.4	75.1	73.4	71.2
LM-softmax	73.6	74.3	72.7	70.5
Focal loss	74.3	74.8	73.6	71.1
Center loss	73.3	74.5	72.8	70.5

to better results. Therefore, how to learn sample weights more smoothly based on the classification performance is the focus of our future work.

5. Conclusion

In this paper, we proposed a novel leader learning mechanism that optimizes both model and loss function through cost-sensitive learning. For neural network classification, zero-one loss is chosen as the leader to optimize the cost-sensitive factor per misclassified sample within general cross entropy (GeCE). GeCE as the executor performs model optimization with good convergence property. To facilitate effective optimization in loss function learning, we incorporated the momentum idea. We conducted experimental simulations on a visible synthetic dataset, ten real-life datasets, MNIST and CIFAR-10/100. The results demonstrate that leader learning still converges sufficiently fast and achieves better classification accuracy when the conventional optimization algorithm based on fixed cross entropy fails to converge. It is empirically concluded that the learned loss function for a specific dataset in one model remains applicable in another model.

CRediT authorship contribution statement

Siyuan Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation. **Linbo Xie:** Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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